

Assignment-IV on JAVA programming

1. WAP in java to create a class `Animal` with attributes like `name` and `age`. Create a subclass `Dog` that inherits from `Animal` and adds a specific attribute like `breed`. Implement methods to display information about both animals and dogs.

Create a class `Puppy` that inherits from `Dog`. Add an attribute specific to puppies, like `milkTeeth`. Implement methods to display information about puppies, including inherited attributes.

In the `Animal` class, create a method `makeSound()`. Override this method in both the `Dog` and `Puppy` classes to produce different sounds. Demonstrate how method overriding works.

2. WAP in java to create an abstract class `Shape` with an abstract method `calculateArea()`. Create concrete classes like `Circle`, `Rectangle`, and `Triangle` that inherit from `Shape` and implement the `calculateArea()` method.
3. WAP in java to create an interface `Drawable` with a method `draw()`. Create classes like `Circle` and `Square` that implement the `Drawable` interface.
4. WAP in java to create an interface `Movable` with methods like `moveUp()`, `moveDown()`, `moveLeft()`, and `moveRight()`. Create another interface `Resizable` that extends `Movable` and adds methods like `resizeWidth()` and `resizeHeight()`. Create a class `Rectangle` that implements `Resizable`.